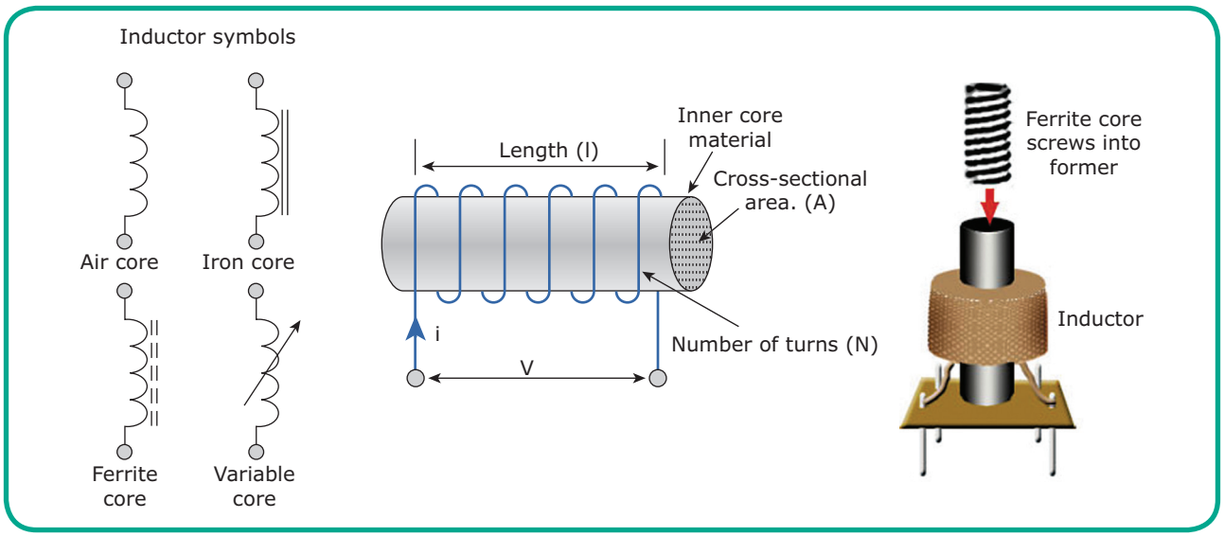


Figure 2.11 Variable Inductors

2.2.4. 'Q' factor (Quality factor) of an inductor

The ratio of the inductive reactance to the effective resistance is known as the 'Q' of an inductor.

$$Q = \frac{2\pi fL}{R}$$

where f is the AC frequency, L is the inductance and R is the resistance.

2.2.5 Chokes

A choke is an inductor that is designed to block unwanted high frequency ac in an electrical circuit. But it allows low frequency and dc.

There are two classes of chokes.

1. Power Chokes
2. RF Chokes

Power and audio frequency chokes typically have an iron core to increase their inductance and make them more effective filters. RF chokes use ferrite core, which operate effectively at high frequencies and is shown in Figure 2.12.

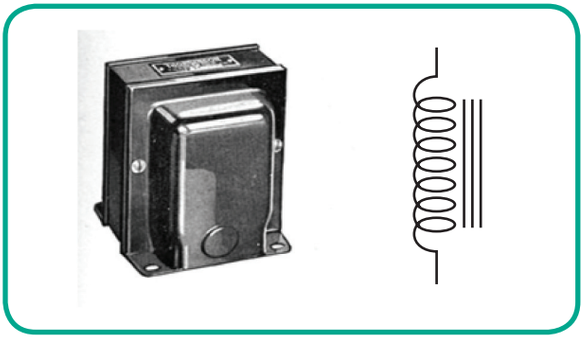


Figure 2.12 Chokes

2.2.6 Relay

A relay is an electrically operated switch and is shown in Figure 2.13. The current flowing through the coil of a relay creates a magnetic field which attracts the lever and changes the switch contacts. Relay is used in applications to turn ON and OFF a circuit by a low power signal.

Uses: Relays are used in stabilizers, UPS (Uninterrupted power supply) and three phase change over switches.

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